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Surface Roughness Effects On Flow Over Aerofoils

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ABSTRACT

An experimental investigation on the effect of surface roughness on a NACA 0012 aerofoil is performed. The experiments include measurements for the pressure distribution, drag coefficients, and lift coefficients. Two different rough surfaces (P120 and P80) are analyzed and their data are compared to those over a smooth surface. Measurements are obtained for angles of attack in the range of $-15°$ to $30°$ with $3°$ steps and for two Reynolds numbers; 1.7×10^5 and 3.4×10^5. Also, measurements of the velocity and turbulence intensity profiles are collected for the flow around the aerofoil at various pre-chosen stations. Laser measurements were performed for two surfaces; smooth and rough (P80), and for two angles of attack; $0°$ and $15°$. Roughness is shown to have a large effect on the flow over the aerofoil. The minimum value for the drag coefficient increases with increasing roughness size on the surface. It causes a reduction in lift over angles of attack from $0°$ up to around $10°$. In the stall region (beyond $\alpha = 10°$), the lift of the rough surface is seen to increase over the corresponding value for the smooth surface, with a higher increase for the higher roughness size. The separation mechanism is also affected significantly by the presence of surface roughness, where roughness is found to delay the separation. Further, the aerofoil efficiency is shown to decrease with roughness.

SYMBOLS

Latin symbols

c	chord length of the aerofoil
C_p	pressure coefficient, $Cp = 2(P - P_\infty) / \rho U^2_\infty$
C_D	drag coefficient, $C_D = 2D / \rho U^2_\infty S$
C_L	lift coefficient, $C_L = 2L / \rho U^2_\infty S$
D	drag force
L	lift force
P	static pressure
P80	rough surface with roughness grade size of 80 μm
P120	rough surface with roughness grade size of 120 μm
r	distance normal to the aerofoil surface
Re	Chord Reynolds number
S	plan area of the aerofoil
U_∞	free stream velocity
U	horizontal velocity component around aerofoil
x	axial distance from the aerofoil leading edge

Greek symbols

α	angle of attack
δ	boundary layer thickness
ρ	density of air

Subscript

e	local external flow value
rms	root-mean-square value of the fluctuating velocity
∞	free stream value

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1. INTRODUCTION

The objective of this work is to provide a better understanding of the effect of surface roughness on flow over aerofoils. Analysis of flow over an aerofoil has many practical applications in addition to its academic interest. The importance of the effect of roughness on aerofoils in aviation work is well realized. Also the performance of axial flow compressors is largely dependent on the surface of the stationary cascade of aerofoils[1].

Surface roughness is a common feature of most operating aerofoil surfaces. Roughness may arise from the manufacturing process, long period of service, and/or natural accumulations (ice, snow, frost, salt spray). Taylor's findings[2] indicate the average roughness height on in-service gas turbine blades may range from about 1.5 μm to about 10 μm. This is very rough considering that the thickness of the boundary layer is on the order of 1 mm. Freezing rain and ice, because of their roughness and the shape change produced on the aerofoil, cause a large number of general-aviation accidents each year. According to Luers[3], it was reported that in 1979, icing was blamed for 49 general aviation accidents. Solutions to icing, such as anti-icing and deicing are expensive and not always adequate for different reasons. Recently, Paula[4] reported that 35 air-carrier incidents worldwide have been attributed to inadequate deicing. Ice affects a plane's performance by hampering the wings' lift capacity. Even in cases where accumulation of ice is not severe, it still forces the aircraft toward more fuel consumption to compensate for the loss of lift.

Roughness, in general, has a large effect on heat transfer and fluid dynamics. Despite some previous experimental and numerical work where surface roughness and pressure gradient are involved, the information that has been obtained on this subject still remains far from complete. Also the process of boundary-layer separation in the presence of surface roughness is not fully understood.

For zero-pressure-gradient flows, as shown by Hosni *et al.*[5], the presence of roughness may increase the heat transfer coefficient and the friction factor by up to 40 percent. For accelerated flows, as shown in[6], a rough wall behaves differently from a smooth wall. For the smooth wall, acceleration causes a decrease in the relative turbulence level and a decrease in Stanton number. On the other hand, accelerated rough-wall boundary layers have two competing effects. The acceleration causes a decrease in the relative turbulence level, but it also causes the boundary layer to become thinner and the skin friction to increase. This, in turn, results in a nondimensionally larger roughness element and larger roughness effect.

It was recognized that surface roughness can not only promote boundary-layer transition from laminar to turbulent, but also affect significantly the subsequent flow development at Reynolds numbers well beyond the critical value. Bloch and Mueller[7] reported experimental data on rough aerofoils at low Reynolds number and they concluded that distributed roughness shifted the separation bubbles forward, thus changing the aerodynamic characteristics of the aerofoil. The performance of the aerofoil was also found to be affected by the distributed roughness elements. Chakroun *et al.*[8] presented drag and lift coefficients for a NACA 2412 flapped aerofoil for different surface roughnesses at a chord Reynolds number of 300,000. For zero flap angle, they reported that roughness causes a reduction in drag and lift. Such reduction was observed to increase with the increase of the roughness size of the aerofoil. However, the maximum efficiency of the aerofoil (max lift to drag ratio) was found to increase with the roughness size.

Several unique applications utilize aerofoils at chord Reynolds numbers below 5×10^5 such as: remotely piloted vehicles, gliders, high-altitude vehicles, wind turbines and propellers; [9] and [10]. This paper presents the results of an experimental investigation undertaken to clarify the influence of roughness on symmetrical aerofoils. Measurements of pressure, lift and drag coefficients were made for angles of attack in the range of –15° to 30° with 3° steps. The data was collected for three different roughness models (smooth, rough-P120, and rough-P80) and at two Reynolds numbers; 1.7×10^5 and 3.4×10^5. Also, measurements of the velocity and turbulence intensity profiles were made for the flow around the aerofoil at some pre-chosen stations. Laser measurements were performed for a smooth surface and for a rough surface of P80. Two angles of attack were studied during the laser measurements; 0° and 15°. The Reynolds number at which the laser measurements were collected was 1.7×10^5.

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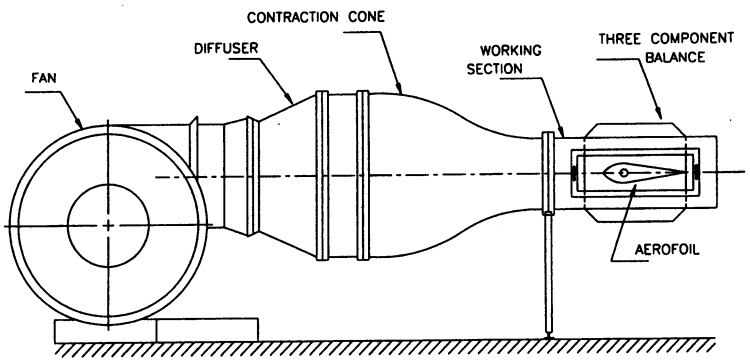


Figure 1. Schematic diagram of the blower tunnel.

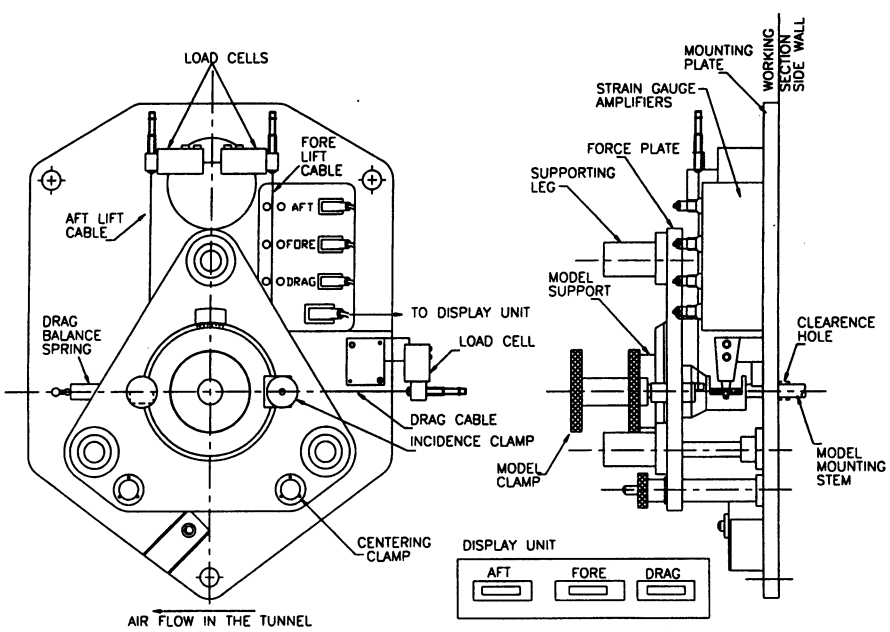


Figure 2. Schematic diagram of the three-component balance.

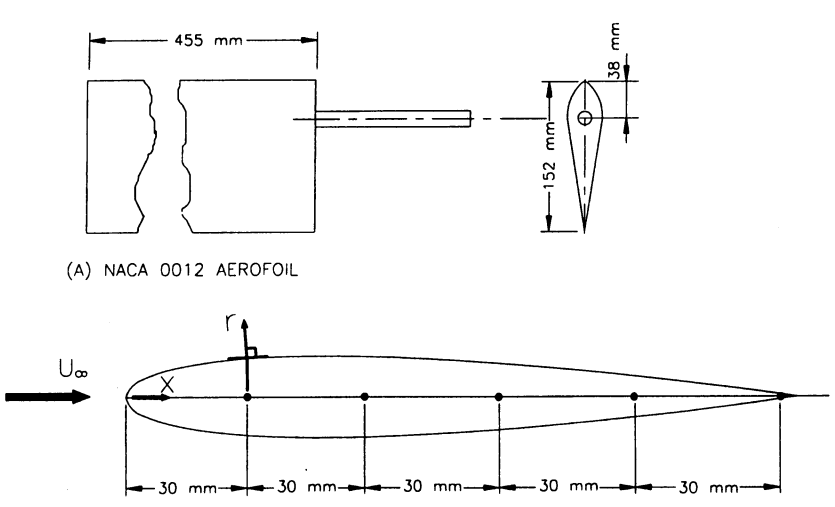


Figure 3. Schematic diagram of NACA 0012 symmetrical aerofoil.

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2. EXPERIMENTAL SET-UP AND PROCEDURE

2.1 Wind tunnel

The experiments were conducted in a blower tunnel. A schematic diagram of the tunnel is shown in Figure 1. The tunnel is an open return type where the open section is discharged directly to the atmosphere, so the pressure in the test section is nearly atmospheric. The working section has internal dimensions of 456 mm × 457 mm. Sockets are provided for mounting a three-component balance on the working section to measure lift and drag forces, and static pressure tappings at the entrance to and exit from the contraction permit the air velocity in the working section to be calculated.

The maximum air speed in the working section is 33 m/s. An investigation at the central plane of the working section using a hot-wire anemometer showed that the r.m.s. value of the axial velocity fluctuations in the working section was not greater than 0.5 percent of the mean velocity, indicating an acceptable level of turbulence.

2.2 Drag, lift and pressure measurements

The electronic TE81/7661 three-component balance (shown in Figure 2) was used to measure the drag and lift forces; this is by measuring the axial and the vertical forces exerted by the air stream. The forces were transmitted by way of flexible cables to strain-gage load cells which measured, respectively, the fore and aft lift forces and the drag force. Pressure measurements were obtained by using an aerofoil with 23 pressure tappings. The pressure tappings of the aerofoil were connected to an electronic micromanometer, where the pressure readings of both upper and lower surfaces were recorded in mm water.

The aerofoil used was the NACA 0012 symmetrical aerofoil. It has a chord length of 152 mm and a span width of 455 mm. The incidence angle of the aerofoil varied from –15° to 30° in steps of 3°. A schematic drawing of the NACA 0012 aerofoil is shown in Figure 3(a). The data was collected for the three different roughness models mentioned before.

The surface roughness was achieved by affixing a sand paper on the surface using two-sided adhesive tape. Before gluing the sand paper, it was soaked in water for a few hours to make it flexible and to avoid formation of sharp edges on the surface of the sand paper. The thickness of the papers including the grid size and the adhesive tape for the rough surface (P80) and the rough surface (P120) are 0.5 mm and 0.4 mm respectively. The roughness was determined based on the grid size of the sand paper. This is not necessarily the equivalent sand-grain roughness. Equivalent sand grain roughness was defined by Schlichting[11] as a hydrodynamic length scale and not a physical length scale. In fact, Schlichting showed that two different sands with the same physical scale can have different equivalent sand grain values.

2.3 LDA measurements

LDA measurement stations for the aerofoil are shown in Figure 3(b). Velocity and turbulence intensity profiles for the smooth and the rough surface were measured for two angles of attack; 0° and 15°. For the 15° angle of attack, measurements were made for the upper and lower surface flows. The LDA used is a 2D Dantec 60X FiberFlow system with a 60 mm probe. The FiberFlow system consists of a 300 mW 2-color Argon-Ion model 5500A laser system (Ion Laser Technology), Dantec's 60X modular optics and a mounting bench, where the optics are configured in a back-scatter mode. The probe used is a 60X61 model with a beam separation distance of 38 mm and a front lens of 400.7 mm focal length. The measuring volume formed by the two pairs of laser beams is an ellipsoid of minor axes of about 116 μm each and a major axis of about 2.44 mm. In the experimental arrangement, the minor axes were parallel to the vertical measurement plane and, consequently, the major axis was perpendicular to the measurement plane. A Dantec LDA signal processor (58N40 Enhanced Flow Velocity Analyzer) is used to extract velocity information after the signal exits the receiving optics. The probe is mounted on a Dantec 41T30 rotational support to allow rotation in the vertical plane. Both the probe and the probe support are, in turn, mounted on a domestic-made support which allows rotation in a horizontal plane. By doing so, the normally inaccessible positions (mostly near the wall) can be readily reached. The probe, held by both probe supports, is mounted on a Dantec 3D lightweight traversing system with 57G15 traversing interface frame. The travers-

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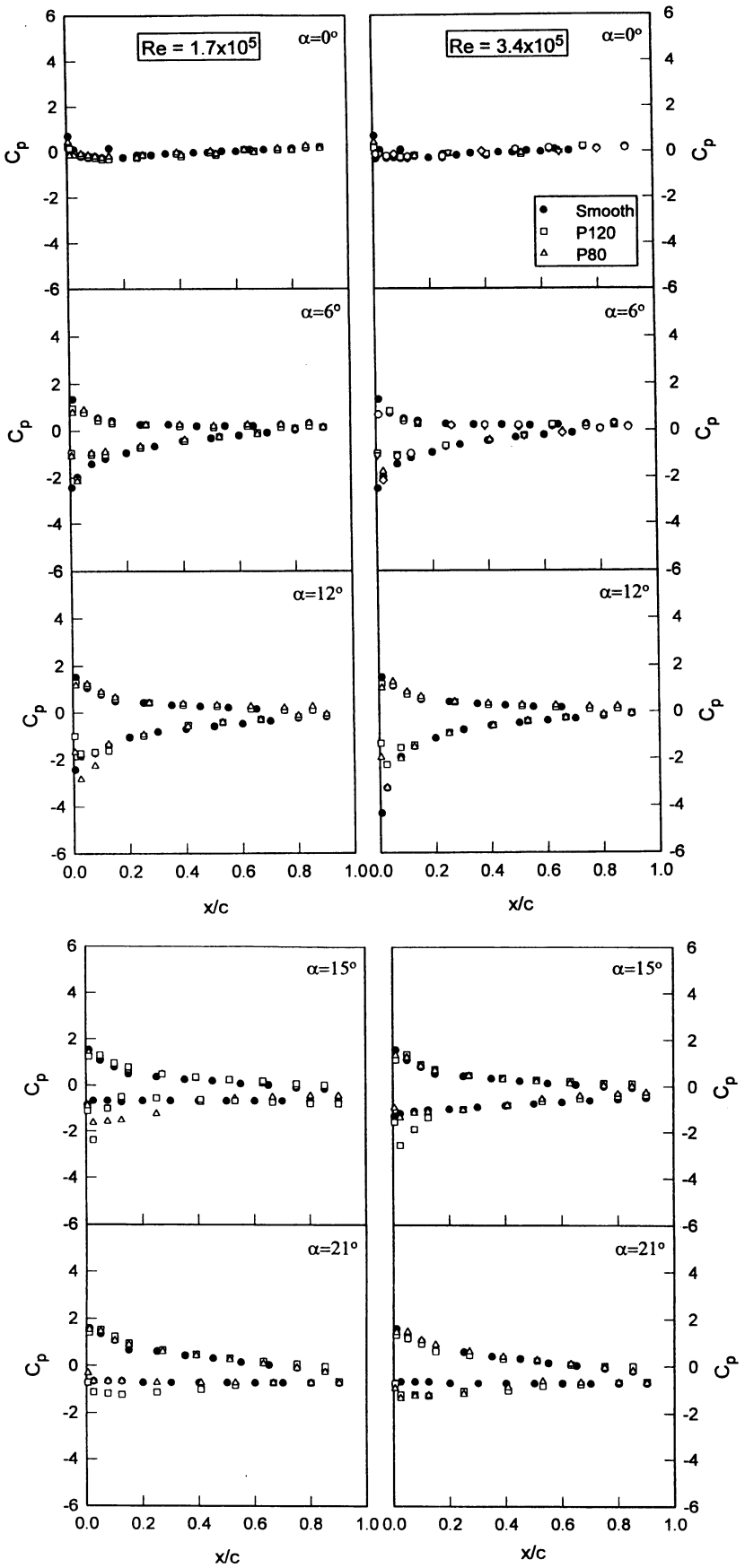


Figure 4. Surface pressure distribution for smooth and rough aerofoils at chord Reynolds numbers of 170,000 and 340,000.

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ing system has a high precision of 10 μm and maximum travels of 590, 590 and 680 mm in the x, y and z directions respectively. It is interfaced to the computer where the movements can be controlled by the keyboard or by the FLOware program. FLOware, a Dantec modular-oriented software, is used to control the experiment, move the traverse, create files, acquire and process the data, and calculate the required statistical quantities of the flow.

The source for the signal of the LDA system is the seeding particles. A smoke generator using Shell Onida Oil is used to seed the room. In order to provide uniform spatial seeding for the measurements, the room was seeded and filled with smoke prior to the test and the blower was used to distribute the particles uniformly. The smoke generator is then turned off before collecting the data.

2.4 Data acquisition of LDA signals

For each run, the flow in the wind tunnel was established and the LDA system was activated. The aerofoil was carefully cleaned before each smooth run. Measurements were taken for four stations, 30 mm apart in the axial direction. For each station, measurement traverses were made in the r-direction (the normal direction to the local aerofoil surface). For each radial traverse, the probe was rotated by a certain angle in the horizontal plane and/or vertical plane when necessary. In fact, these rotations were necessary for two reasons; preventing the laser beams from being obstructed by the aerofoil body, and helping in taking measurements very close to the surface of the aerofoil.

After establishing the starting point, a pre-programmed traverse file for the traverse system was loaded and the data acquisition variables (e.g. number of samples, time-out, etc) were fixed. Once all the data is acquired, the probe is moved to the starting point and the raw data files are used for processing and obtaining the required flow quantities.

3. EXPERIMENTAL RESULTS AND DISCUSSION

A test on a symmetrical aerofoil with a smooth surface and two different rough surfaces gives an introduction to the understanding of the effect of roughness on aerofoils. The measurements were obtained for fifteen angles of attack in the range of –15° to 30° in steps of 3°. The measurements were collected for two Reynolds numbers; namely $Re = 1.7 \times 10^5$ and 3.4×10^5. The Reynolds number was calculated based on the chord length of the aerofoil. The measurements are presented in terms of the drag, lift, and pressure coefficients.

In addition, laser measurements of the mean velocity and turbulence intensity profiles were made for the smooth surface and the rough surface of P80 at $Re = 1.7 \times 10^5$, and at two angles of attack of 0° and 15°.

The uncertainty in the experimental data is estimated based on the ANSI/ASME Standard on Measurement Uncertainty[12] following the procedures of Coleman and Steele[13]. For the data of C_D, C_L, and C_p, the overall uncertainty ranges from ± 1% to ± 2%, depending on the flow conditions. For the laser measurements, the overall uncertainties in the mean and rms velocities were estimated to be ± 2% and ± 4% respectively. Therefore, data collected for different aerofoil surfaces can be compared with each other with a good degree of confidence.

Figure 4 presents the measured pressure distribution on the aerofoil in the form of the pressure coefficient (C_p) versus the normalized axial distance from the aerofoil leading edge (x/c) at some representative angles of attack of 0°, 6°, 12°, 12°, 15°, and 21°. The pressure distribution measurements were obtained for the lower and the upper surfaces of the smooth and the two rough aerofoils at two Reynolds numbers; $Re = 1.7 \times 10^5$ and $Re = 3.4 \times 10^5$. The difference between the pressure distribution of the lower and the upper surfaces is maximum at the leading edge and minimum close to the trailing edge. For zero angle of attack, as expected, the pressure difference is nearly zero. As the angle of attack increases, the difference starts to increase until it reaches maximum at $\alpha = 12°$. It is observed from the Figure that when the angle of attack increases beyond a value of 12°, there is an increase in the pressure difference for both rough aerofoils over that of the smooth aerofoil, particularly near the leading edge. However, no appreciable effect for the roughness on the pressure difference is seen for angles of attack smaller than $\approx 12°$. For $\alpha = 21°$ and higher, the pressure on the upper surface of the aerofoil shows a uniform distribution, which is typical in the stall region. Figure 4 reveals that the sensitivity of the pressure distribution measurements to the

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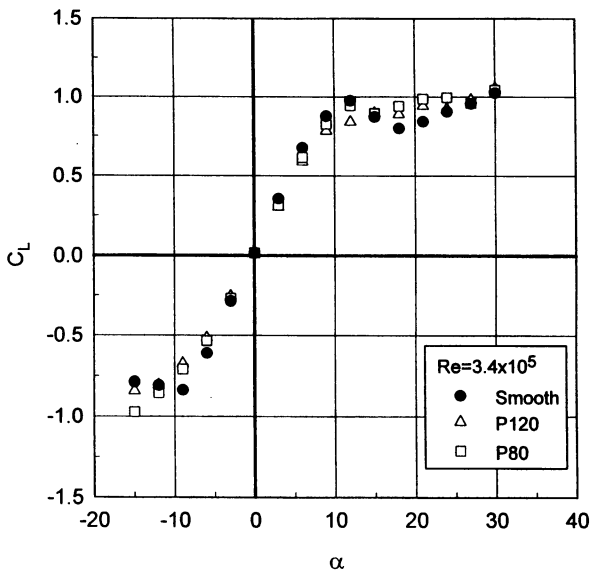


Figure 5. Variation of lift coefficient with angle of attack for the smooth and rough surfaces at a chord Reynolds number of 340,000.

Reynolds numbers is not noticeable over the Reynolds numbers considered in the present study.

Figure 5 presents the lift coefficient data as a function of the angle of attack for the three surface models (smooth, rough-P120 and rough-P80) for a Reynolds number of 3.4×10^5. It is to be mentioned at this point that the lift data of $Re = 1.7 \times 10^5$ is almost the same as those for $Re = 3.4 \times 10^5$, and therefore is not given. Figure 5 shows that the zero lift angle of attack is not affected by roughness, having a value of 0° for all surfaces as would be expected for a symmetrical aerofoil. Maximum lift for the smooth aerofoil is obtained at about 12° angle of attack with a maximum lift coefficients of about 1. This maximum value marks the beginning of the stall region which is characterized by a sharp decrease in the lift up to $\alpha = 18°$, after which the lift starts to increase again in a gradual manner. In both rough surfaces considered, it is seen that maximum lift was reduced and accompanied by a gradual stall. The rough surface P120 appears to cause a more gradual stall than the rough surface P80. In the stall region, for $\alpha \geq 15°$, the lift is seen to be higher for both rough surfaces compared to the smooth surface, with relatively higher lift produced by the surface of higher degree of roughness (P80). Since all the error in the data are believed to be bias error, the data can be compared with each other with a high degree of confidence.

Figure 6 shows the variation in the drag coefficient versus the angle of attack for the same three surfaces and Reynolds number as those for Figure 5. It is seen from the Figure that the drag coefficient is minimum at the zero angle of attack. The effect of roughness on the minimum drag is observed, where the minimum drag is seen to increase with the roughness, taking the values ≈ 0.001, 0.025 and 0.03 for the smooth surface, P120 rough surface and P80 rough surface respectively. The drag coefficient increases gradually until it reaches a maximum value at the maximum angle of attack. The surface roughness effect is seen to increase the drag. Such increase gets higher with the degree of roughness of the aerofoil surface.

Experimental data expressed as lift-to-drag ratio versus angle of attack is given in Figure 7. The ratio C_L/C_D is usually regarded as a measure of the aerofoil efficiency. The Figure shows that the maximum value for all surfaces is reached at an angle of attack of 6°, with maximum C_L/C_D ratios of 45.5, 20 and 13 for the smooth surface, rough surface (P120) and rough surface (P80) respectively. The roughness reduces the aerofoil efficiency up to the stall angle of 12°, with a more reduced efficiency for the larger roughness size. Compared with the smooth surface, the maximum decrease in the maximum lift-to-drag ratio due to the surface roughness is 55%. The effect of Reynolds number, within the range covered in the present work, is negligible.

The LDA results are shown in Figures 8 through 13. Velocity profiles were measured at

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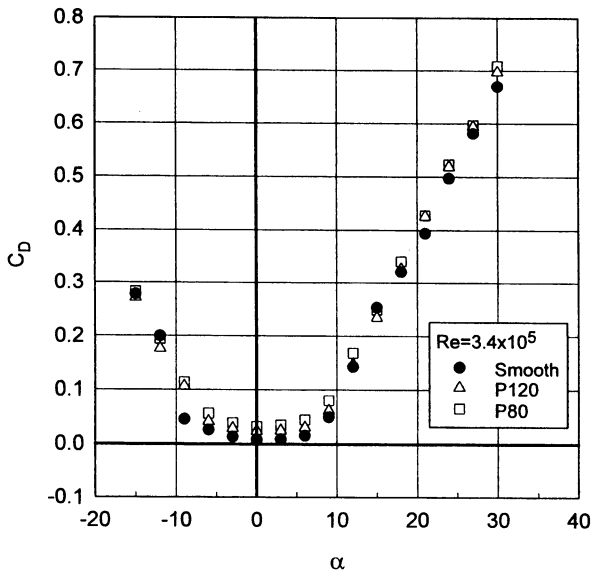


Figure 6. Variation of drag coefficient with angle of attack for the smooth and rough surfaces at a chord Reynolds number of 340,000.

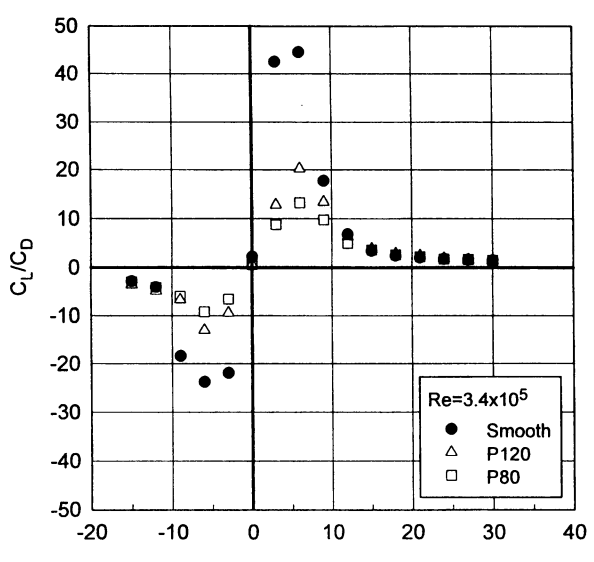


Figure 7. Variation of lift-to-drag ratio with angle of attack for the smooth and rough surfaces at a chord Reynolds number of 340,000.

x/c = 0.2, 0.4, 0.6, and 0.8. Figure 8 shows the gradual progression of the horizontal longitudinal velocity profiles for both surfaces at zero angle of attack. For the smooth surface separation starts to occur at x/c = 0.4, however at 0.6 it appears to have disappeared and then reappeared at 0.8. It is worth mentioning at this point that the experiments of Figure 8 were repeated twice to ensure that the data is free from errors. The boundary layer on a low Reynolds number aerofoil often remains laminar well into the adverse pressure gradient region. In many cases, laminar separation takes place with transition to turbulence occurring within the separated shear layer. After transition, mixing causes the turbulent free shear layer to reattach and develop as a turbulent boundary layer. For the rough surface (P80), it is seen that the boundary layer is developing and thickening in the downstream direction, and no sign of separation is observed. The corresponding turbulence intensity profiles are given in Figure 9. Surprisingly, the smooth surface displays lower turbulence in spite of flow separation.

Figure 10 shows the velocity profiles for the lower surface of the aerofoil at an angle of

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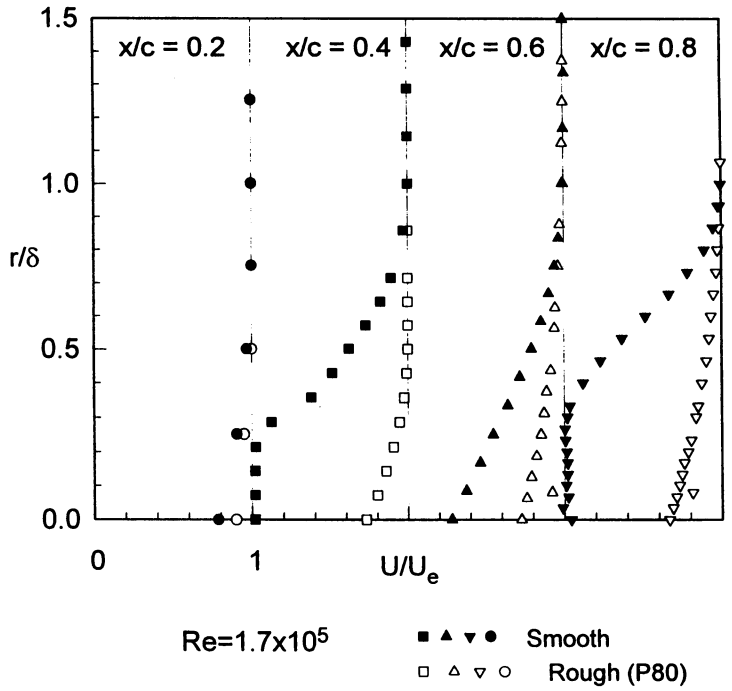


Figure 8. Horizontal component of longitudinal mean velocity profiles for zero angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000.

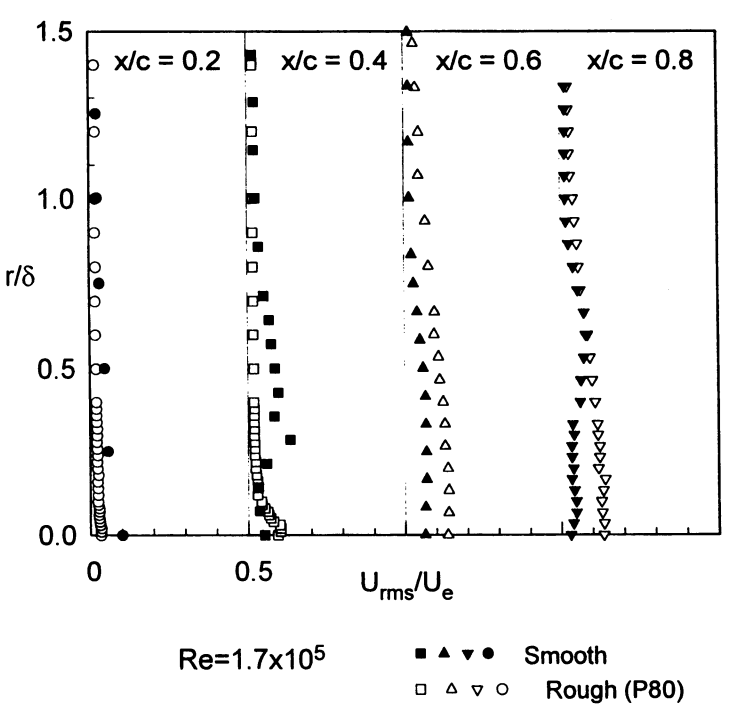


Figure 9. Horizontal component of longitudinal turbulence intensity profiles for zero angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000.

attack of 15°. Flow over both surfaces shows no sign of separation. The smooth surface, however, is shown to thicken the boundary layer. This is presumably a laminar boundary layer. The corresponding turbulence intensity is given in Figure 11. The turbulence intensity is high, if a laminar boundary layer is assumed to exist, which in such a case may be attributed to flow

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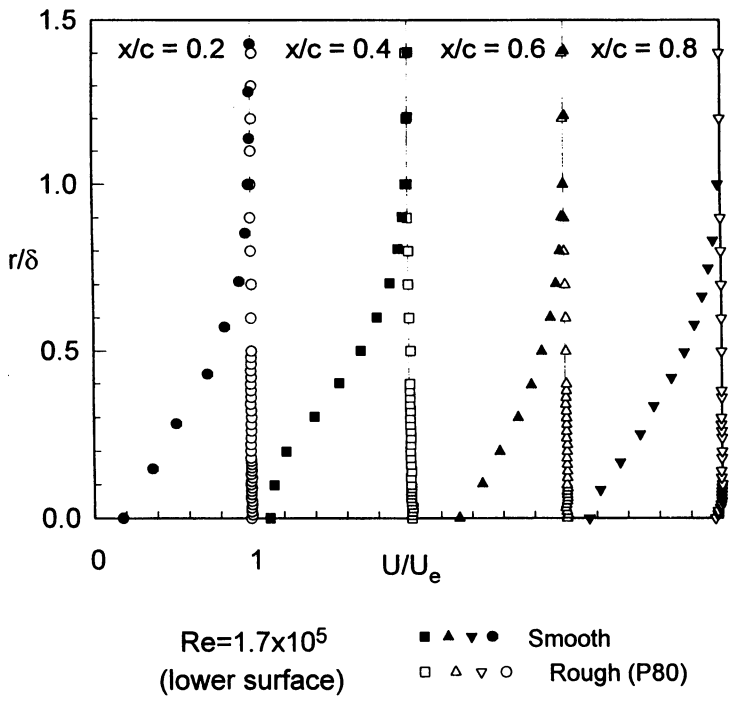


Figure 10. Horizontal component of longitudinal mean velocity profiles for 15° angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000 (Lower surface).

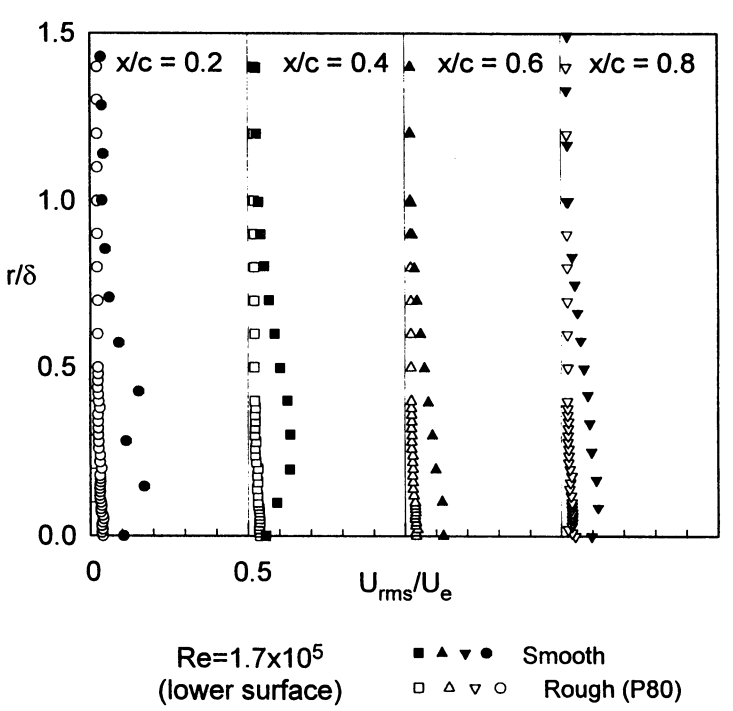


Figure 11. Horizontal component of longitudinal turbulence intensity profiles for 15° angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000 (Lower surface).

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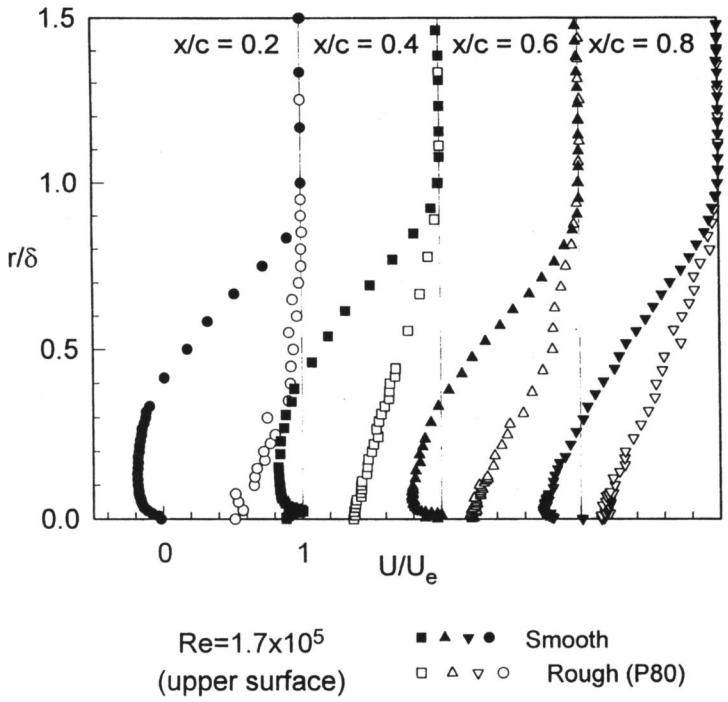


Figure 12. Horizontal component of longitudinal mean velocity profiles for 15° angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000 (Upper surface).

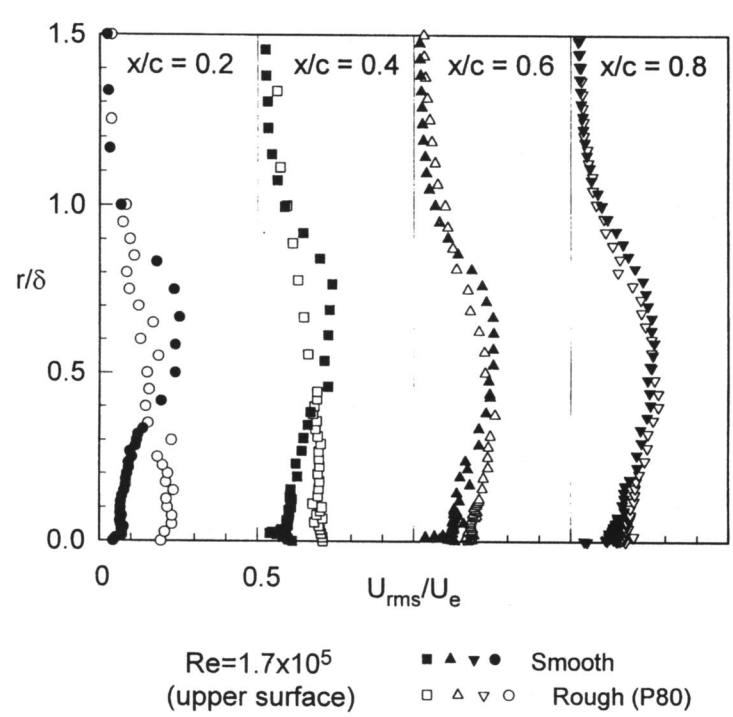


Figure 13. Horizontal component of longitudinal turbulence intensity profiles for 15° angle of attack, at four x/c stations, for smooth and rough surfaces, at a chord Reynolds number of 170,000 (Upper surface).

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unsteadiness such as intermittent or transition flow conditions. For the same angle of attack of 15°, the velocity profiles for the upper surface are presented in Figure 12. While the flow over the rough surface shows a slight tendency towards separation, the smooth-surface flow does separate over the entire aerofoil. It is to be noted that the angle of attack of 15° is well within the stall region. It is also worth mentioning here that for the rough surface (P80) the stall was gentle, while it was sharp for the smooth surface. Turbulence intensity profiles for the upper surface of 15° angle of attack are given in Figure 13.

In brief, the surface roughness is shown to influence the aerodynamic behavior of the aerofoil. It seems that roughness delays separation of the flow. It is known that turbulent flows that are influenced by surface roughness are usually divided into three regions: aerodynamically-smooth flows which are those flows where the roughness is so small that the flow behaves as if the wall was smooth, fully-rough flows where the roughness so dominates the momentum transport to the wall that viscous effects are negligible, and traditionally-rough flows which occur at Reynolds numbers in between, and both viscous and roughness effects are significant. This division of turbulent flows may explain the delayed separation for the rough-surface flow of the present study.

4. SUMMARY AND CONCLUSIONS

An experimental study of the effect of surface roughness on flow over NACA 0012 symmetrical aerofoil has been undertaken. The study included measurements of the pressure, drag, and lift coefficients, in addition to LDA velocity and turbulence intensity profiles. Two different rough surfaces were used (P120 and P80) where the data is compared to those for a smooth surface. Measurements were taken for two Reynolds numbers of 1.7×10^5 and 3.4×10^5.

Surface roughness has a large effect on the flow over an aerofoil. The pressure difference between the lower and upper surfaces of the aerofoil increases with surface roughness, particularly in the stall region. The minimum value of the drag coefficient increases with increasing roughness size on the surface. The separation of the flow is also affected by the surface roughness, where roughness is shown to delay separation.

Roughness changes the flow characteristics around the aerofoil. It causes a gradual increase in drag. A small decrease in lift is seen before the stall; however, this decrease is within the uncertainty of the data. In the stall region, roughness is seen to increase the lift. The increase in drag coefficient varies in the range of 5 to 30 percent, and the increase in lift in the stall region reaches a maximum of 15 percent. The zero lift angle of attack was not influenced by surface roughness. The angle of attack where the aerofoil can have its maximum efficiency or maximum lift-to-drag ratio does not seem to change with the size of roughness on the surface, for the roughness sizes in the present work. Efficiency of the aerofoil was decreased by a maximum of 55% due to the roughness.

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